

## General information

The day-ahead market takes place once per day. Bids are submitted at midday of **day D-1** for the 24 hours of **day D**. Participants must submit their bids separately for each hour.

Each bid comprises:

- The **amount of energy (GWh)** they want to sell or buy
- The **price (€/MWh)** at which they are willing to do so

### Market Clearing Process

For each hour, generation bids are ordered from the lowest to the highest price, forming the **supply curve**, while demand bids are ordered from the highest to the lowest price, forming the **demand curve**. The intersection of these two curves determines the **market clearing price**, the amount of **energy traded**, and the **generation schedule** for that hour.

### Uniform-Price Auction

The day-ahead market is a **uniform-price** auction, which means that all accepted generators are paid the same price and all buyers pay the same price. This price corresponds to the market clearing price. In theory, generators maximise their profits by bidding their opportunity cost (that is, their avoidable production cost), since this increases the probability of being dispatched while ensuring that their costs are covered.

### Imbalances

The day-ahead market fixes revenues and payments **one day in advance (D-1)**. However, real production and consumption during **day D** may differ from scheduled values for several reasons. These deviations are called **imbalances**, and they lead to **penalties**.

## Game Description

In this game, you will act as a generator participating in the day-ahead market. The exercise consists of four rounds, and each round represents one specific hour of a different day. The rounds are played sequentially, and after each round the results will be made public before the next round begins. In every round, you must submit a bid and aim to maximise your profit.

### Public information

There are five generators, each representing one technology:

- Nuclear: 8 GW
- Gas: 20 GW
- Coal: 10 GW
- Hydro: 15 GW
- Solar-Wind: 20 GW

At the beginning of each round, the Market Operator will provide an **estimate of total demand** for that hour. In addition, relevant market information will be published, including the **hydro reserves**, the **CO<sub>2</sub> price**, and any other data necessary to support your bidding decisions.